Leading TEXHEC #1

They told me Golang wasn't for games.
"The Garbage Collector will kill your frame rates," they said.
I decided to build a massive-scale RTS from scratch anyway.

Meet TEXHEC the project I've been leading and building for over a year.
It's not just a game, it's a technical challenge to find order in endless chaos (seemingly game world).

What is TEXHEC?

Imagine a world where you have to choose between automation and manual care for everything.
A Real-Time Strategy game where: - Scale is King: We are simulating 1,000,000 tiles with thousands of buildings and units—all in real-time. - Total Control: From technological dominance to total annihilation, the player has to conquer a seemingly endless world. - Complexity simplified: Complex tactics meet a high-performance engine built to handle them without breaking a sweat.

Choice of golang

Most developers shy away from Go for high-performance rendering.
We embraced it.
By following Data-Oriented Design (DOD) and building our own ECS (Entity Component System) framework from scratch,
we've turned the "GC problem" into a non-issue.

The results?

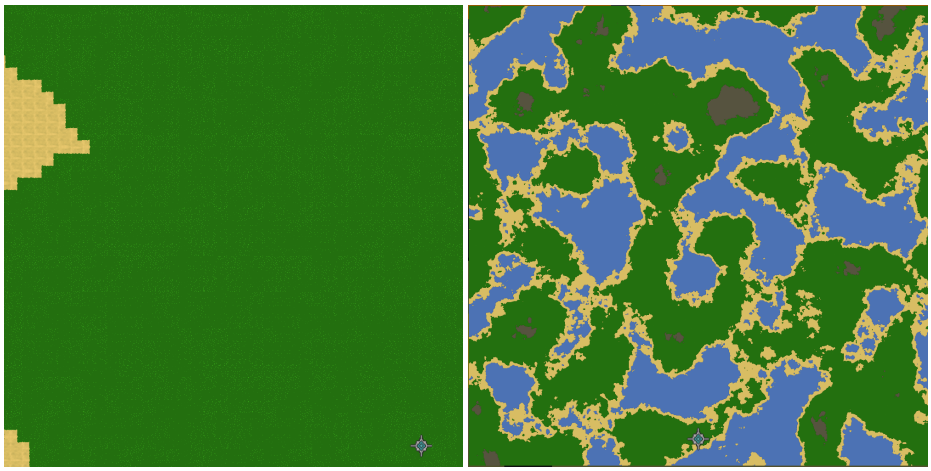
A massive map generated in seconds and rendered in less than 6ms—even on a 5-year-old Intel i5 with integrated graphics.

Why build from scratch?

In a world of bloated engines, TEXHEC is lean. We use less than a dozen dependencies. - Simplicity = Performance. - No decades of technical debt. - Total architectural freedom. Total control over architecture allows us to choose our mental models and how hardware executes them.

Scale

Here see the scale of the map with photo of whole map and of bottom right map corner:



See zooming map under this link

Follow for more

Over the next few weeks, I'll be sharing a series of posts titled "Leading TEXHEC," where I'll dive into the unique modules I've written—from custom Assets and Hierarchy systems to how we handle real-time recording and state replication.

The journey from a blank `main.go` to a simulated world has been a wild ride. I'm excited to finally show you what's happening behind the scenes. What's your take on using Go for gamedev? Let's argue (respectfully) in the comments.

See also TEXHEC

#Golang #GameDev #RTS #ECS #TEXHEC